Fiscal Year:	FY 2015	**Task Last Updated:**	FY 05/14/2015
PI Name:	Hammond, Timothy M.D.		
Project Title:	Yeast Colony Survival in Microgravity Depends on Ammonia Mediated Metabolic Adaptation and Cell Differentiation -Flight		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology		
Space Biology Cross-Element Discipline:	(1) Reproductive Biology		
Space Biology Special Category:	None		
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Comments:	Alternate email: Timothy.Hammond@va.gov NOTE: PI previously at Tulane University until 2007 (Ed., 09/2010) Joint appointments Professor, Department of Internal Medicine, Nephrology Division, Duke University School of Medicine, and Staff Physician, Durham VA Health Care System.		
Project Type:	Flight	**Solicitation / Funding Source:**	2012 Space Biology NNH12ZTT001N
Start Date:	08/01/2013	**End Date:**	07/31/2017
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:	Taylor, Elizabeth	**Contact Phone:**	650.604.1783
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Flight Program:	ISS		
Flight Assignment:	Rapid Turn Around Flight ISS NOTE: End date changed to 7/31/2017 per NSSC information (Ed., 7/25/16) NOTE: End date is 7/31/2016 (originally 7/31/2015) per NSSC information (Ed., 3/27/14)		
Key Personnel Changes/Previous PI:	May 2015 report: Added Drs. Corey Nislow and Guri Giaever as Co-Is.		
COI Name (Institution):	Birdsall, Holly M.D., Ph.D. (Baylor College of Medicine) Nislow, Corey Ph.D. (University of British Columbia) Giaever, Guri Ph.D. (University of British Columbia)		
Grant/Contract No.:	NNX13AN32G		
Performance Goal No.:			

Performance Goal Text:	
Task Description:	Yeast chemical genomics is a powerful tool that is being applied to drug discovery. This approach allows the study of compound-target relationships in the context of a living cell with a systems biology view that includes both the cell target and the interdependent networks that respond to chemical stress. The modulation of yeast colonies, especially the redox potential, allows study of different drug pathways. This application investigates microbial metabolic adaptation and cell differentiation to the multiple stimuli encountered in space flight environments, using the yeast Saccharomyces cerevisiae as a model system. We have shown that microgravity activates a unique set of genes in yeast. Our previous work has largely defined yeast cellular stress defense response to microgravity, but stress defense responses are only involved in the survival of yeast in liquid cultures. It has recently been found that survival of plate-based yeast colonies is dependent on signaling mediated through ammonia which drives the multicellular yeast colony to differentiate into two populations, L or lower cells and U or upper cells. These two populations have different metabolic pathways and establish a cycle of nutrients that allows the U cells to continue to proliferate at the leading edge. Without ammonia signaling, the entire colony dies. Lack of convection in space is likely to dramatically impact signaling by ammonia, a volatile gas. We postulate that the flow of nutrient and ammonia signals will differ in microgravity where there is little to no convection. We hypothesize that yeast colony survival in microgravity depends on ammonia mediated metabolic adaptation and cell differentiation rather than on stress defense. To test our hypothesis, we will identify the mechanisms mediating colony stratification into U and L cells under multiple gravitational environments: microgravity emulated by Rotating Wall Vessel and Random Positioning Device, static one g cultures, and ultimately microgravity of space flight. Our experimental model will compare liquid and solid phase yeast cultures and make use of available mutant strains of yeast with deletion of the enzymes involved in ammonia signaling and the connected alternative metabolism activation, enzymes involved in stress responses, and the Msn4 promoter and SSA4 gene that we have previously identified as mediating response to microgravity. Relevant metabolic pathways will be studied using green fluorescent protein-linked reporter genes that can be analyzed by confocal microscopy of intact colonies, and flow cytometry of cells divided into L and U populations by sucrose density centrifugation. There are similarities between multicellular yeast colonies and tumors of clusters of mammalian cells in tissue culture. Thus insights into the effect of microgravity on cell to cell signaling may have far-reaching applications. Microgravity could provide a physiologically responsive test bed for drug pathway analysis at diverse redox states.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Yeast chemical genomics is a powerful tool that is being applied to drug discovery. This approach allows the study of compound target relationships in the context of a living cell with a systems biology view that includes both the cell target and the interdependent networks that respond to chemical stress. It has recently been found that survival of plate-based yeast colonies is dependent on signaling mediated through ammonia which drives the multicellular yeast colony to differentiate into two populations, L or lower cells and U or upper cells. Lack of convection in space is likely to dramatically impact signaling by ammonia, a volatile gas. There are similarities between multicellular yeast colonies and tumors of clusters of mammalian cells in tissue culture. Thus insights into the effect of microgravity on cell to cell signaling may have far-reaching applications. Microgravity could provide a physiologically responsive test bed for drug pathway analysis at diverse redox states.
Task Progress:	Our hypothesis to be tested is: Microgravity-induced changes in yeast are induced, at least in part, through changes in density driven ammonia convection and the subsequent effects on Msn4- and Sfp1-dependent genetic and cellular differentiation of giant yeast colonies. Specific Aim 1: To evaluate the role of ammonia-based signaling in the cellular and genetic differentiation of giant yeast colonies under simulated microgravity conditions. We will utilize a series of yeast strains with normal, partial, or complete deletion of ammonia production genes combined with eGFP reporters to quantify expression of genes associated with giant colony differentiation. These experiments will compare effects of static conditions versus microgravity simulation by clinorotation on giant yeast colonies. Specific Aim 2: To determine the role of Msn4 and Sfp1 in the cellular and genetic differentiation of giant yeast colonies with changes in ammonia production under simulated microgravity conditions. We will utilize a series of yeast strains with normal and mutated Msn4 and Sfp1 promoter regions, together with ammonia production defects, to determine the effects on gene expression. These will be combined with eGFP to quantify expression of genes associated with giant colony differentiation. These experiments will compare effects of static conditions versus microgravity simulation by clinorotation on giant yeast colonies. Specific Aim 3: To confirm and extend the results obtained in our ground-based studies on the changes in ammonia dependent Msn4 and Sfp1 mediated genetic and cellular differentiation of giant yeast colonies in the true microgravity environment. Utilizing giant yeast colonies grown on agar plates, we will conduct parallel flight studies that can be activated and preserved for subsequent analysis using cold storage. This work aims to use space-based tools to demonstrate the basis of a unique redox state reflecting the electrical state mediating drug metabolism in cancers. Space would therefore provide a powerful clinical tool to understand drug pathways in a manner not currently demonstrable by any means. Task Progress: Summary - The science verification test is successful and the mission is ready to fly. - Conditions including timing, selection of background genes to delete, and analytical dye choices and concentrations have been fully optimized. Hardware - The Bioserve Space Technologies Handware hardware is available, facile to operate, and ready to fly.

	Analytical - Analytical techniques have been optimized, validated, and documented.
Bibliography Type:	Description: (Last Updated: 03/31/2026)
Abstracts for Journals and Proceedings	Nislow C, Lee AY, Allen PL, Giaever G, Gebbia M, Stodieck L, Hammond JS, Birdsall HH, Hammond TG. "Fitness Profiling of Yeast Mutants Reveal Unique Genetic Responses to Spaceflight." Presented at 30th Annual Meeting of the American Society for Gravitational and Space Research, Pasadena, CA, October 22-26, 2014. 30th Annual Meeting of the American Society for Gravitational and Space Research, Pasadena, CA, October 22-26, 2014. https://www.asgsr.org/index.php/presentation-abstracts-10-26-2014 ; search for Abstract ; accessed 5/21/15. , Oct-2014
Abstracts for Journals and Proceedings	Hammond T. "Microbes, Microgravity and Microvirulence." 3rd Annual International Space Station (ISS) Research and Development Conference, Chicago, Illinois, June 17-19, 2014. 3rd Annual International Space Station (ISS) Research and Development Conference, Chicago, Illinois, June 17-19, 2014. , Jun-2014
Articles in Peer-reviewed Journals	Nislow C, Lee AY, Allen PL, Giaever G, Smith A, Gebbia M, Stodieck L, Hammond JS, Birdsall HH, Hammond TG. "Genes required for survival in microgravity revealed by genome-wide yeast deletion collections cultured during spaceflight." Biomed Res Int. 2015;2015:976458. Published online 2015 Jan 13. PubMed PMID: 25667933 ; PubMed Central PMCID: PMC4309212 ; http://dx.doi.org/10.1155/2015/976458 , Jan-2015
Awards	Nislow C. (Corey Nislow, Host Faculty member - Timothy Hammond, Awardee Fellow.) "Awardee Fellow, Peter Wall Institute of Advanced Studies Visiting International Research Scholar, May 2015.." May-2015
Books/Book Chapters	Abdelmoaty H, Hammond TG, Wilson BL, Birdsall HH, Clement JQ. "Identification of putative major space genes using genome-wide literature data." in "Biotechnology." Ed. D. Ekinci. Intech, Open Access, April 2015. p. 93-116. ISBN 978-953-51-2040-7, http://dx.doi.org/10.5772/60412 , Apr-2015